

Database-Traceable Uncertainty Quantification for Layered-Tissue RF Dosimetry

1 Traceable inputs



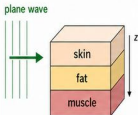
- 112 tissue entries
- 104 with complete dielectric data
- conductivity uncertainty: conformal interval



- thermal uncertainty: documented min/max



2 Reproducible model

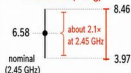


Transmission-line SAR
+
1D Pennes bioheat

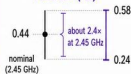
50 W/m²
0.9 / 2.45 / 5.8 GHz

3 Propagated bounds

Peak SAR (W/kg)



Peak ΔT (°C)



2.45 GHz example:

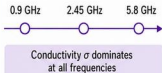
peak SAR 6.58 [3.97, 8.46] W/kg;

peak ΔT 0.44 [0.24, 0.58] °C

Cross-frequency widening across
0.9–5.8 GHz is about 2–4×.

4 Dominant drivers

A. Peak SAR



B. Peak ΔT



Documented tissue-property uncertainty changes SAR and temperature margins; the controlling driver depends on endpoint and frequency.